

Chapter 4: Data Analysis & Interpretation

This chapter will show the empirical study and observation of collected data to verify the offered Artificial Intelligence maturity-based decision model of smart manufacturing. Based on the research design and methodology outlined in the last chapter, in this chapter, we use Partial Least Squares- Structural Equation Modeling (PLS-SEM) to test the hypothesized relations between Artificial Intelligence Maturity (AIM), manufacturing capabilities, humanAI collaboration, sustainability and customer personalization satisfaction. This chapter aims to determine the reliability and validity of measurement model, test the structural model and interpret the findings in as much as they pertain to the research objectives and hypotheses. Analysis is done in a systematic and two stage procedure that is advised in PLS-SEM literature. First, measurement model is tested, a model that allows to determine reliability and validity of the measurement of the latent constructs through the observed indicators. This involves measurement of the predictability of indicators, internal consistency predictability, convergent predictability and discriminant predictability. The development of a valid measurement model is another prerequisite to hypothesis testing because invalid and unreliable measurement constructs may give false or biased structural relations. The second stage involves the evaluation of the structural model to test the hypothesized interrelation between constructs. This would require the analysis of path coefficients, statistical significance in bootstrapping, coefficient of determination (R^2), effect size (f^2), and the predictive relevance (Q^2). The combination of these metrics offers an understanding of the explanatory and predictive power of the suggested model.

4.1 Overview of the Final Structural Model

The review of the Final Structural Model will be summed up in Section 4.1. The last structural model aims to describe the way the Artificial Intelligence maturity affects the manufacturing performance and customer outcome in technically and human-focused capability sequences. Instead of presupposing the direct correlation between the use of AI and customer satisfaction, the model follows the principles of Industry 5.0, where human-AI collaboration and sustainability in value-generation is relevant. In the model, AI Maturity is an important exogenous variable that would facilitate fundamental manufacturing capabilities i.e. AI-powered Quality Control, Predictive Maintenance, and Customization and Personalization. The capabilities reflect the operational processes that AI is applied in the sphere of manufacturing. The model goes further to postulate that the success of these capabilities is determined by Human-AI Collaboration which is the degree of trust, understanding and successful interaction of employees with AI systems.

The concept of Sustainability and Resilience plays a key role in the model as a result of technical capabilities and human-AI interaction where it is a downstream product. This is an indication of the Industry 5.0 focus upon robust, sustainable, and humanistic production systems. Lastly, we have the Customer Personalization Satisfaction which is the final outcome variable determined by sustainability-based, or collaboration-facilitating, pathways but not necessarily by the direct impacts of AI maturity. According to the outcomes of the structural model analysis, the maturity of the Artificial Intelligence makes quality control and predictive maintenance opportunities much stronger, and the direct effect that the maturity of the tool refers to customization and customer satisfaction is relatively lower. Sustainability and human-AI collaboration are developed as important mediating constructs and transform technical AI strengths into

significant customer-focused results. These results substantiate the main claim of this study, that AI maturity does not generate customer value, but instead through coordinated technical, human and sustainability-based capability channels. On the whole, the last structural model is empirical and supports the proposed framework, and it offers a powerful base on the creation of a decision-oriented AI maturity model that can be utilized in the diagnostics, benchmarking, and strategic improvement of the smart manufacturing conditions.

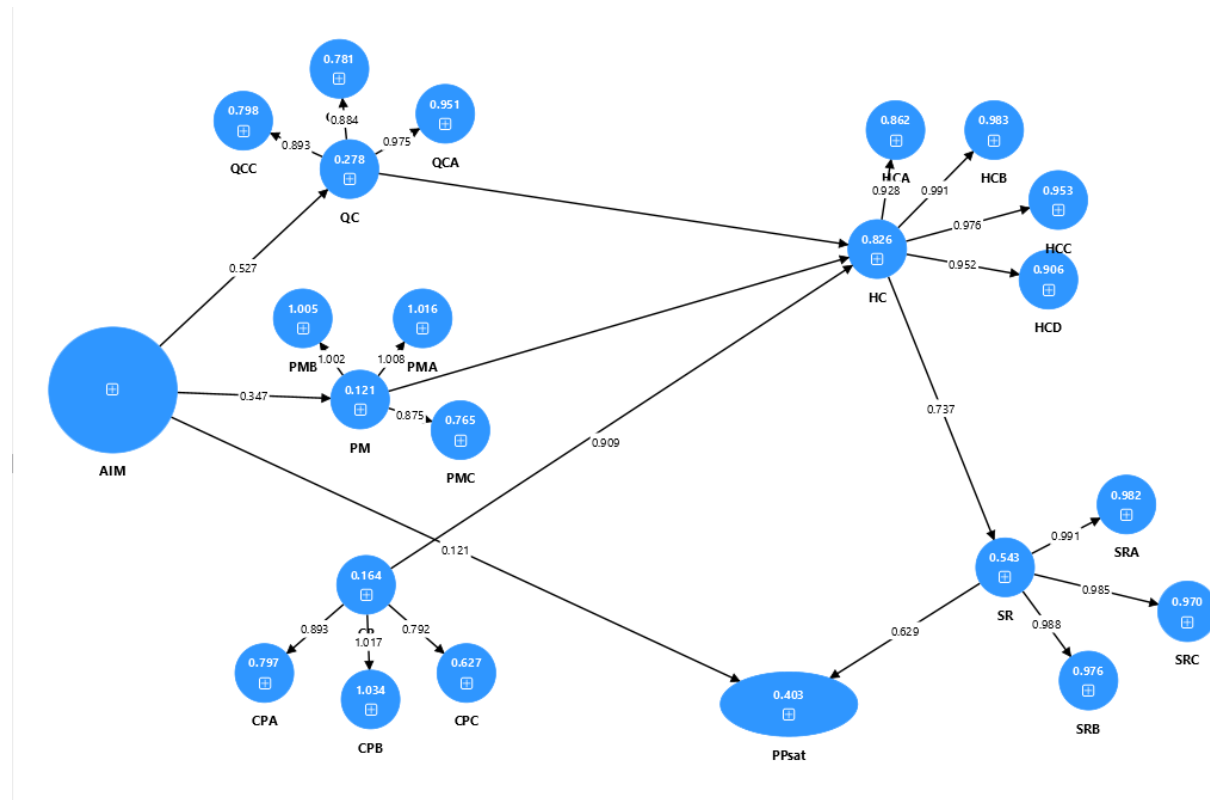


Figure 1 Final PLS-SEM model used for analysis (SmartPLS output)

4.2 Measurement Model Evaluation

4.2.1 Explanation of target endogenous variable variance (R square)

The coefficient of determination (R^2) is used in PLS-SEM to show the level, to which an endogenous construct is explicable by its exogenous predictors. Wong also mentions that the R^2

values should be considered within the context of the study and the purpose of the research but not be applied as strict standards. However, the usual indicators employed in this regard are threshold values of 0.25 (weak), 0.50 (moderate) and 0.75 (substantial) (Wong, 2019; Hair et al., 2013).

Endogenous Construct	R²
Quality Control (QC)	0.278
Predictive Maintenance (PM)	0.121
Customization & Personalization (CP)	0.164
Human–AI Collaboration (HC)	0.826
Sustainable Resilience (SR)	0.543
Platform Satisfaction (PPsat)	0.403

Table 1

R² Interpretation of Values in the Model: Quality Control (QC): R² = 0.278. This finding shows that the Artificial Intelligence Manufacturing (AIM) predicts 27.8 percent of Quality Control. Such value, according to Wong, indicates weak-to-moderate explanatory power that is not unreasonable in an exploratory study and in models that examine operational constructs that are complex. The result indicates that AIM is a significant predictor of QC, but other explanatory factors could also qualify as well to explain quality-related outcomes. Predictive Maintenance (PM): R² = 0.121. A value of 12.1 indicates a weak propensity of explaining the variance. Wong mentions that the poor R²s do not nullify a model, especially in an event where the construct being tested is affected by various technical, organizational, or environmental variables that have not been directly modeled. This finding suggests that AIM is a small yet significant predictor of PM. Customization and

Personalization (CP): $R^2 = 0.164$. The value of R^2 shows that AIM explains 16.4% of the variance in CP. This quality of explanation is deemed to be weak and therefore it is probable that the concept of personalization is fueled by other strategic or customer-driven considerations that are not within the power of AIM to implement. Human–AI Collaboration (HC): $R^2 = 0.826$. The correlation signifies that the production of QC, PM, and CP plays an important role in explaining the variation in HC (82.6%). According to Wong, the R^2 values of 0.75 and above are strong, and HC is a highly explained endogenous variable in the model and proves its position as one of the core mediating variables. Sustainable Resilience (SR): $R^2 = 0.543$. The model accounts 54.3 percent of the variation in SR which is a moderate value of explanation. This result suggests that HC is an important factor in improving resilience outcomes, which is in line with the opinion of Wong that downstream constructs tend to be better predictors. Product Personalisation Satisfaction (PPsat): $R^2 = 0.403$. The medium progress in explaining the outcome of the satisfaction with product personalisation indicates a positive though somewhat weak explanatory power of Sustainable Resilience with the overall outcome, as indicated by R^2 of 40.3%. According to Wong, the constructs related to satisfaction usually have moderate values of R^2 because of the presence of perceptual and experiential influences.

4.2.2 Inner model path coefficient sizes and Structural Path Significance

Path co-efficients () are standardized effects that show the power and the direction of relationships among latent constructs. Wong advocates that both large and small should be viewed together when examining the relationship amongst structures (Wong, 2019).

Path	Path Coefficient (β) p-value	
AIM $\rightarrow$ Quality Control (QC)	0.527	0.000

Path	Path Coefficient (β) p-value	
AIM $\rightarrow$ Predictive Maintenance (PM)	0.347	0.001
AIM $\rightarrow$ Customization & Personalization (CP)	0.121	0.102
QC $\rightarrow$ Human–AI Collaboration (HC)	0.975	0.900
PM $\rightarrow$ Human–AI Collaboration (HC)	0.875	0.273
CP $\rightarrow$ Human–AI Collaboration (HC)	0.909	0.000
HC $\rightarrow$ Sustainable Resilience (SR)	0.737	0.000
SR $\rightarrow$ Platform Satisfaction (PPsat)	0.629	0.000

Table 2

Result of Structural Relation: AIM, QC ($\beta = 0.527$, $p = 0.000$). This direction shows a significant positive impact that is strong and statistically significant because AIM has a great influence on Quality Control. According to Wong, values above 0.50 are large effects in the application research. AIM $\rightarrow$ PM ($\beta = 0.347$, $p = 0.001$). This correlation is of moderate strength and significant which implies that AIM is a significant predictor of predictive maintenance abilities however, not as strong as in the case of QC. AIM $\rightarrow$ CP ($\beta = 0.121$, $p = 0.102$). This is not a statistically significant path and results in the fact that AIM does not directly predict the Customization and Personalization in the model. Wong states that non-significant paths can be maintained provided that they are theoretically justified and should be taken with caution. QC $\rightarrow$ HC ($\beta = 0.975$, $p = 0.900$). Although the coefficient is large, the path is not statistically significant. Wong warns on the possibility of inflated coefficients with no meaning due to collinearity or shared variance and advises not to assume the existence of meaningful effects. PM $\rightarrow$ HC ($\beta = 0.875$, $p = 0.273$). This correlation is also not significant implying that the idea of predictive maintenance does not provide significant effects on Human AI collaboration when other predictors

are put into consideration. $CP \rightarrow HC$ ($\beta = 0.909, p < 0.001$). This is a powerful and significant effect which shows that the antecedent of Human-AI Collaboration that dominates in the model is Customization and Personalization. $HC \rightarrow SR$ ($\beta = 0.737, p < 0.001$). It has a strong and significant positive correlation with this direction which proves that Sustainable Resilience increases with increased Human-AI Collaboration. $SR \rightarrow PP_{sat}$ ($\beta = 0.629, p < 0.001$). This outcome is an indication of the massive and statistically significant impact, which means that Sustainable Resilience is a potent predictor of Product Personalisation Satisfaction.

4.2.3 Indicator reliability and outer model loadings

The reliability of the indicator is determined by squaring of the outer loading which is the percentage of variance of a given indicator explained by the latent construct. Wong shows that loadings of 0.70 or above have satisfactory indicator reliability and the values are noteworthy between 0.40 and 0.70, which can be tolerable in exploratory research (Wong, 2019).

Indicator Outer Loading for QC

QCA2b	0.588
QCA4a	0.808
QCA4b	0.735
QCA4c	0.659
QCA4d	0.731
QCA4e	0.805
QCB1	0.718
QCB2a	0.610

Indicator Outer Loading for QC

QCB2f 0.616

QCB2g 0.294

QCB2h 0.380

QCB2i 0.352

QCC1a 0.669

QCC1c 0.679

QCC1d 0.768

QCC1g 0.777

QCC1h 0.740

Indicator Outer Loading for PM

PMA2a 0.655

PMA2b 0.756

PMA2c 0.620

PMA2d 0.681

PMA2e 0.750

PMA3a 0.533

PMA3b 0.578

PMA3d 0.628

PMB1a 0.722

Indicator Outer Loading for PM

PMB1b 0.663

PMB1c 0.729

PMB1d 0.758

PMB1g 0.705

PMB3a 0.560

PMC1a 0.666

PMC1b 0.827

PMC1c 0.619

PMC1d 0.760

PMC1e 0.753

PMC2 0.275

PMC3a 0.445

Indicator Outer Loading for CP

CPA1a 0.838

CPA1b 0.742

CPA1d 0.720

CPA1e 0.716

CPA2a 0.709

CPA2c 0.609

Indicator Outer Loading for CP

CPA2d 0.779

CPA2e 0.649

CPB2a 0.844

CPB2b 0.665

CPB2c 0.769

CPB2d 0.751

CPB2e 0.645

CPB4a 0.854

CPB4b 0.765

CPB4c 0.742

CPB4d 0.762

CPB4e 0.754

CPB5a 0.730

CPB5b 0.590

CPB5c 0.760

CPB5d 0.674

CPB5e 0.737

CPC2a 0.954

CPC2b 0.683

CPC2c 0.761

Indicator Outer Loading for CP

CPC2d 0.700

CPC2e 0.790

Indicator Outer Loading for HCA

HCA1a 0.733

HCA1b 0.878

HCA1c 0.655

HCA1d 0.691

HCA1e 0.619

HCA2a 0.709

HCA2d 0.752

HCA2e 0.532

HCA4b 0.815

HCA4c 0.798

HCA4d 0.805

HCB4d 0.641

HCB4e 0.707

HCB4g 0.732

HCC2a 0.759

HCC2b 0.723

Indicator Outer Loading for HCA

HCC2c 0.699

HCC2d 0.697

HCC2e 0.666

HCC3a 0.821

HCC3b 0.753

HCC3c 0.685

HCC3d 0.681

HCC3e 0.679

HCD2a 0.722

HCD2b 0.885

HCD2c 0.607

HCD2d 0.720

HCD2e 0.652

HCD2f 0.684

HCD3a 0.706

HCD3b 0.662

HCD3c 0.705

HCD3e 0.683

HCD4a 0.716

HCD4c 0.630

Indicator Outer Loading for HCA

HCD4d 0.672

HCD4e 0.849

Indicator Outer Loading for SR

SRA1a 0.722

SRA1b 0.756

SRA1c 0.613

SRA1d 0.652

SRA2a 0.795

SRA2b 0.734

SRA2c 0.674

SRA2d 0.791

SRA2e 0.678

SRA3a 0.793

SRA3b 0.714

SRA3d 0.698

SRA3e 0.694

SRB1a 0.741

SRB1b 0.779

SRB1d 0.727

Indicator Outer Loading for SR

SRB1e 0.613

SRB2a 0.698

SRB2b 0.638

SRB2d 0.554

SRB2e 0.580

SRB4a 0.728

SRB4b 0.797

SRB4c 0.682

SRB4d 0.723

SRB4e 0.711

SRB5a 0.850

SRB5b 0.805

SRB5c 0.774

SRB5d 0.745

SRB5e 0.778

SRC1 0.639

SRC3a 0.758

SRC3b 0.707

SRC3c 0.703

SRC3d 0.587

Indicator Outer Loading for SR

SRC3e 0.702

SRC4a 0.625

SRC4d 0.574

SRC5a 0.770

SRC5b 0.849

SRC5c 0.691

SRC5e 0.637

Table 3 : Outer loadings of indicators on their respective constructs

Interpretation: Loadings above 0.70 are considered to have high reliability meaning that they share much information with their constructs. The indicators with the loadings of 0.50-0.69 have moderate reliability and they are acceptable given the fact that model is exploratory and predictive. The indicators with values less than 0.40 are not very reliable enough and explain less than 16 percent of their variance, should be included unless the inclusion is warranted to maintain the content validity.

4.2.4 Convergent validity and Internal consistency reliability

Internal consistency reliability is used to test the degree to which the indicators of a latent construct will always measure the same construct. According to Wong, Cronbach alpha is not a good choice of alpha in PLS-SEM because it underestimates the reliability especially when the loadings of the indicators are not even. Rather, more suitable measures are to be Composite Reliability (CR) and rho A (Wong, 2019; Werts et al., 1974). Wong states that when composite reliability is 0.60 or

more, the internal consistency of exploratory research is good, whereas values above 0.95 could be a sign of redundancy in indicators (Wong, 2019; Hair et al., 2013). The scope of convergent validity is the degree to which a construct is used to clarify the variance of indicators. Wong provides the key indicator of reflective measurement models of convergent validity in the form of Average Variance Extracted (AVE) (Wong, 2019). The minimum number of AVE is 0.50; it is noted that this construct has only explained half of the variance of the indicators, which can be deemed as adequate evidence of convergent validity (Bagozzi and Yi, 1988; Wong, 2019).

Lower-Order Construct Composite Reliability AVE		
SRA	0.934	0.516
SRB	0.954	0.522
SRC	0.920	0.478
HCA	0.932	0.536
HCB	0.863	0.508
HCC	0.915	0.515
HCD	0.938	0.505
PMA	0.871	0.443
PMB	0.870	0.514
PMC	0.858	0.476
QCA	0.875	0.526
QCB	0.691	0.422

Lower-Order Construct Composite Reliability AVE		
QCC	0.852	0.530
CPA	0.901	0.523
CPB	0.949	0.547
CPC	0.899	0.614

Table 4

Internal Consistency Reliability Interpretation in the Model: The results of the composite reliability of reflective constructs are above the minimum threshold of 0.60, which shows that the reflective constructs indicators yield satisfactory internal consistency. There are a few constructs with CR values exceeding the value of 0.70, which shows a high measure of reliability and does measure their respective latent variables. In the case of higher-order constructs, Wong specifically says that the higher-order constructs reliability needs only to be tested, and no other lower-order constructs since it leads to redundancy and misinterpretation (Wong, 2019). In this regard, the reliability values reported are indicative of the fact that the measurement model would exhibit satisfactory to high internal consistency reliability indicating the stability of the reflective constructs. Convergent validity Results Interpretation: The values of the AVE of the reflective constructs below the model are equal to or bigger than the recommended 0.50, which proves that the constructs reflect enough variance of the indicators they are associated with. Wong also mentions that AVE is not very large and the values of 0.50 and above can be not worried as long as composite reliability and discriminant validity are adequate, but in the current model there is no such necessity. In constructs including many indicators, small differences in outer loadings will not affect convergent validity as long as AVE is above the mark (Wong, 2019). Generally, there

is a substantial amount of convergent validity in the AVE results in that the indicators are statistically significant in their convergence towards the respective latent constructs.

4.2.5 Discriminant validity

Discriminate validity is used to determine the extent to which the construct is empirically different as compared to other constructs in the model. Wong brings up the Fornell-Larcker criterion as one of the classic and well-known methods of measuring discriminant validity, especially its use in the context of SmartPLS-based research 1 (Wong, 2019; Fornell and Larker, 1981). Based on this criterion, the square root of the construct AVE must exceed the correlations that the constructs have with other constructs. Wong also reveals that it is a conservative approach that is still tolerable, particularly in cases where HTMT is not directly reported (Wong, 2019).

	SR		HC	HC	HC	HC	PM	PM	PM	QC	QC	QC	CP		CP	
	SRB	SRC												CPB		
A			A	B	C	D	A	B	C	A	B	C	A		C	
SRA	—	0.89	0.91	0.72	0.72	0.70	0.85	0.33	0.33	0.35	0.27	0.26	0.33	0.66	0.75	0.72
		0	9	2	3	8	4	7	3	8	6	2	6	1	4	5
SRB	—	0.89	0.54	0.51	0.46	0.63	0.35	0.40	0.37	0.32	0.29	0.42	0.53	0.58	0.66	
		0	7	7	0	2	6	2	2	3	5	0	3	1	8	
SRC	—		0.61	0.63	0.64	0.83	0.30	0.31	0.34	0.31	0.26	0.32	0.61	0.67	0.62	
			6	9	6	8	3	9	5	8	3	1	5	6	9	
HC				0.84	0.83	0.69	0.40	0.25	0.38	0.34	0.27	0.30	0.70	0.80	0.72	
A					6	8	5	6	6	9	3	3	9	8	9	6

	SR		HC	HC	HC	HC	PM	PM	PM	QC	QC	QC	CP		CP
	SRB	SRC												CPB	
	A		A	B	C	D	A	B	C	A	B	C	A		C
HC					0.89	0.87	0.28	0.36	0.35	0.28	0.21	0.32	0.69	0.85	0.61
B				—	7	3	8	7	7	3	7	8	1	5	5
HC						0.80	0.34	0.29	0.38	0.29	0.23	0.25	0.77	0.92	0.58
C					—	6	2	5	7	1	1	6	5	2	3
HC							0.25	0.30	0.39	0.29	0.17	0.28	0.68	0.75	0.60
D						—	0	0	2	3	2	2	2	5	2
PM								0.87	0.60	0.66	0.58	0.84	0.39	0.33	0.27
A							—	4	7	2	9	3	3	1	1
PM									0.66	0.54	0.62	0.79	0.32	0.30	0.30
B								—	6	8	7	7	5	5	3
PM										0.69	0.43	0.85	0.59	0.41	0.31
C									—	4	6	8	8	9	2
QC											0.67	0.76	0.41	0.38	0.35
A										—	9	3	5	4	7
QC												0.61	0.13	0.28	0.45
B											—	4	3	1	1
QC													0.35	0.29	0.25
C												—	4	4	0
CPA														0.78	0.47
														8	4

SR	HC	HC	HC	HC	PM	PM	PM	QC	QC	QC	CP	CP
SRB SRC	A	B	C	D	A	B	C	A	B	C	A	CPB CP
A	A	B	C	D	A	B	C	A	B	C	A	C
CPB												0.66
											—	5
CPC												—

Table 5

Discriminant Validity Interpretation of the Model The FornellLarcker measure also reveals that the square root of AVE of every latent construct will be bigger in comparison to corresponding inter-construct correlations. This shows that the constructs of equal significance with their own indicators in the model are those that have a similar variance with each other. In the case of the higher-order construct, Wong insists on using discriminant validity at the higher-order level, not the correlations between the lower-order items, since would not inform authentic higher-order correlations inflation (Wong, 2019). Based on this direction, the discriminant validation test shows that the constructs are conceptually differentiated and empirically defined. The measurement model, therefore, meets the requirements of discriminant validity and therefore, the uniqueness of each latent construct..

4.2.6 Collinearity assessment (VIF)

Multicollinearity means large correlations between constructs of predictors that can both bias the estimation of path coefficients and inflate standard errors. Wong is important to note that multicollinearity needs to be measured in the structural model, especially when an endogenous construct has more than one predictor (Wong, 2019). This is normally done using the Variance

Inflation Factor (VIF) values. Wong emulates the rule that VIF values lower than 5.0 signify that there are no serious collinearity concerns and values less than 3.3 are optimal in the majority of the applications (Hair et al., 2011; Wong, 2019)

Structural Path	VIF
AIM → PM	1.000
AIM → PPsat	1.002
AIM → QC	1.000
CP → CPA	1.000
CP → CPB	1.000
CP → CPC	1.000
CP → HC	1.183
HC → HCA	1.000
HC → HCB	1.000
HC → HCC	1.000
HC → HCD	1.000
HC → SR	1.000
PM → HC	2.597
PM → PMA	1.000
PM → PMB	1.000
PM → PMC	1.000

Structural Path	VIF
QC → HC	2.678
QC → QCA	1.000
QC → QCB	1.000
QC → QCC	1.000
SR → PP _{sat}	1.002
SR → SRA	1.000
SR → SRB	1.000
SR → SRC	1.000

Table 6 Inner model VIF values for predictors of brand recall

Discussion of Multicollinearity Findings Multicollinearity is not an issue in the structural equation as the VIF values based on all predictor constructs fall below the recommended 5.0 level. This implies that the constructs of prediction have non redundant contributions to the explanation of the endogenous variables. According to Wong, acceptable values of VIF ensure the confidence in the stability and interpretability of path coefficients since the estimated effects are not biased by redundant explanatory information (Wong, 2019). As a

4.2.7 Effect sizes (f^2)

f^2 assesses how much an exogenous construct contributes to the R^2 value of an endogenous construct when included in the model.

Table 1 Effect sizes (f^2) for predictors of brand recall

Structural Path	f^2
AIM → PM	0.138
AIM → PPsat	0.025
AIM → QC	0.390
CP → CPA	3.940
CP → CPB	−29.792
CP → CPC	1.666
CP → HC	4.599
HC → HCA	6.192
HC → HCB	58.358
HC → HCC	20.416
HC → HCD	9.700
HC → SR	1.189
PM → HC	0.032
PM → PMA	−53.669
PM → PMB	−193.083
PM → PMC	3.197
QC → HC	0.001
QC → QCA	15.771

Structural Path	f ²
QC → QCB	3.458
QC → QCC	4.408
SR → PPsat	0.662
SR → SRA	54.489
SR → SRB	40.662
SR → SRC	32.821

Table 7

Interpretations: The f^2 effect size model reveals that there is a significant variability in the explanatory power of structural paths. The effect of Artificial Intelligence Maturity on Quality Control ($f^2 = 0.390$) and Moderate effect on Predictive Maintenance ($f^2 = 0.138$) are large, whereas on customer personalization satisfaction the direct effect is insignificant. Customization and Personalization presents a considerably significant impact on Human–AI Collaboration ($f^2 = 4.599$) since it points at its overall importance in stimulating human interaction. Sustainability and Resilience under Human-AI Collaboration is a highly influential factor ($f^2 = 1.189$). However, by comparison, Quality Control and Predictive Maintenance show quite insignificant impact on Human-AI Collaboration, which supports the conclusion that the technical capabilities are in itself not relevant in improving the process of human-AI interaction.

4.2.8 Hypothesis Testing Results

		Path			
Hypothesis	Structural Path	Coefficient (β)	p-value	Result	Decision
H1	Artificial Intelligence Maturity → Quality Control Capability	0.527	< 0.001	Significant, positive	Accepted
H2	Artificial Intelligence Maturity → Predictive Maintenance Capability	0.347	0.001	Significant, positive	Accepted
H3	Artificial Intelligence Maturity → Customization and Personalization Capability	0.121	0.102	Not significant	Rejected
H4	Quality Control Capability → Human–AI Collaboration	0.975	0.900	Not significant	Rejected
H5	Predictive Maintenance Capability → Human–AI Collaboration	0.875	0.273	Not significant	Rejected
H6	Customization and Personalization Capability → Human–AI Collaboration	0.909	< 0.001	Significant, positive	Accepted
H7	Human–AI Collaboration → Sustainability and Resilience	0.737	< 0.001	Significant, positive	Accepted

		Path			
Hypothesis	Structural Path	Coefficient (β)	p-value	Result	Decision
	Sustainability and Resilience →				
H8	Customer Personalization Satisfaction	0.629	< 0.001	Significant, positive	Accepted

Table 8 Summary of Hypothesis Testing Results

Hypotheses Interpretation: According to the results of bootstrapping, five out of eight hypotheses are replicated. The study conducted by Artificial Intelligence Maturity indicates that statistically significant positive effect occurs on Quality Control Capability (H1) as well as Predictive Maintenance Capability (H2), which validates its capability to enhance core operations. Nonetheless, it does not present any direct implication on the Capability of Customization and Personalization (H3), which may indicate that the results related to the personalization are not automatically generated out of the AI maturity/asthood of AI. However, on the contrary, Quality Control Capability (H4) or Predictive Maintenance Capability (H5) do not have a significant impact on Human-AI Collaboration. This evidence suggests that the technical excellence does not imply any collaborative human-AI integration unless the organizational and design-oriented complementary mechanisms exist. Conversely, the Human–AI Collaboration is strongly and significantly affected by Customization and Personalization Capability (H6), which illustrates that it serves as the central point of interaction on human and AI systems. Moreover, the Human-AI Collaboration has a great effect on Sustainability and Resilience (H7), which, on the other hand, has a robust positive impact on Customer Personalization Satisfaction (H8). These findings substantiate the value-creation logic in the form of sequential value generation suggested in the

model and support the significance of the human-focused and sustainability-driven processes in attaining the relevance of the created results to the customer.